Department of Computer Science and Engineering (Data Science)

S.Y.B.Tech. **Sem: IV**

Subject: Computational Methods and Pricing Models Laboratory

Experiment 10

Name: **SAP ID:**

Date:	**Experiment Title:** Implement Cost of Carry Model to determine commodity prices
Aim	To understand the Cost of Carry Model for pricing futures contracts on commodities. To implement the model using Python. To analyze the relationship between spot price and futures price. To study the effect of interest rate and cost of carry on futures pricing.
Software	Python on Google Colab
Theory	The Cost of Carry Model is used to determine the theoretical price of a futures contract based on the relationship between the spot price and the futures price of an asset. The model assumes that the futures price reflects the cost of holding (carrying) the underlying asset until the maturity of the contract. These costs include interest rates, storage costs, and income (like dividends or yield). The formula for the Cost of Carry Model is: $$F = S \times e^{(r-q)\times T}$$ where: ● **F** = Futures price ● **S** = Spot price of the underlying asset ● **r** = Risk-free interest rate (annual) ● **q** = Dividend yield / income generated from the asset (If not given, assume q=0) ● **T** = Time to maturity (in years) ● **e** = Exponential constant (≈ 2.71828) **Interpretation** ● If **F > S**, the market is in **contango** (positive cost of carry). ● If **F < S**, the market is in **backwardation** (negative cost of carry).

<table>
<tr><td></td><td>
Effect of Cost of Carry

Increase in cost of carry (r increases or q decreases):
Futures price increases.
Decrease in cost of carry (r decreases or q increases):
Futures price decreases.

</td></tr>
<tr><td>Implementation</td><td>
Implementation

Step 1: Define Input Parameters

Spot Price (S)
Risk-Free Rate (r)
Dividend Yield (q) (If not given, take q = 0)
Time to Maturity (T)

Step 2: Apply Cost of Carry Formula

import numpy as np
F = S * np.exp((r - q) * T)

Step 3: Display Futures Price

Print the calculated futures price.

Step 4: Analyze Relationship Between Spot and Futures Price

Compare values of S and F and identify market conditions (contango/backwardation).

Step 5: Sensitivity Analysis

Vary risk-free rate (r) from 1% to 10%.
Observe changes in futures price.

Step 6: Plot Graph

Plot Futures Price vs Risk-Free Rate.
Plot Futures Price vs Time to Maturity.

</td></tr>
<tr><td>Conclusion</td><td>(Write your conclusions)</td></tr>
</table>

Lab 10: Cost of Carry Model for Commodity Futures Pricing

Experiment Title

Implement Cost of Carry Model to determine commodity prices

Aim

- To understand the Cost of Carry Model for pricing futures contracts on commodities
- To implement the model using Python
- To analyze the relationship between spot price and futures price
- To study the effect of interest rate and cost of carry on futures pricing

Theory

The **Cost of Carry Model** is used to determine the theoretical price of a futures contract based on the relationship between the spot price and the futures price of an asset. The model assumes that the futures price reflects the cost of holding (carrying) the underlying asset until the maturity of the contract.

Formula

$$F = S \times e^{(r-q)\times T}$$

Where:

- **F** = Futures price
- **S** = Spot price of the underlying asset
- **r** = Risk-free interest rate (annual)
- **q** = Dividend yield / income generated from the asset (If not given, assume q=0)
- **T** = Time to maturity (in years)
- **e** = Exponential constant (≈ 2.71828)

Interpretation

- If **F > S**: The market is in **contango** (positive cost of carry)
- If **F < S**: The market is in **backwardation** (negative cost of carry)

Effect of Cost of Carry

- **Increase in cost of carry** (r increases or q decreases): Futures price increases
- **Decrease in cost of carry** (r decreases or q increases): Futures price decreases

Implementation

Step 1: Import Required Libraries and Define Input Parameters

```
import numpy as np
import matplotlib.pyplot as plt
import pandas as pd

# Step 1: Define Input Parameters
print("="*60)
print("STEP 1: Define Input Parameters")
print("="*60)

# Input parameters for the Cost of Carry Model
S = 100  # Spot Price (Current price of the commodity)
r = 0.05  # Risk-Free Rate (5% annual interest rate)
q = 0.02  # Dividend Yield / Income yield (2% annual)
T = 1    # Time to Maturity (1 year)

print(f"\nInput Parameters:")
print(f"Spot Price (S): ${S:.2f}")
print(f"Risk-Free Rate (r): {r*100:.2f}%")
print(f"Dividend Yield (q): {q*100:.2f}%")
print(f"Time to Maturity (T): {T} year(s)")
print(f"\nCost of Carry (r - q): {(r-q)*100:.2f}%")
```

```
============================================================
STEP 1: Define Input Parameters
============================================================

Input Parameters:
Spot Price (S): $100.00
Risk-Free Rate (r): 5.00%
Dividend Yield (q): 2.00%
Time to Maturity (T): 1 year(s)

Cost of Carry (r - q): 3.00%
```

Step 2: Apply Cost of Carry Formula

```
print("\n" + "="*60)
print("STEP 2: Apply Cost of Carry Formula")
print("="*60)

# Calculate Futures Price using the Cost of Carry Model
# Formula: F = S * e^((r - q) * T)
F = S * np.exp((r - q) * T)

print(f"\nFormula: F = S × e^((r - q) × T)")
print(f"F = {S} × e^(({r} - {q}) × {T})")
```

```
print(f"F = {S} × e^({(r - q)*T:.4f})")
print(f"F = {S} × {np.exp((r - q)*T):.6f}")
print(f"\nCalculated Futures Price (F): ${F:.2f}")
```

```
============================================================
STEP 2: Apply Cost of Carry Formula
============================================================

Formula: F = S × e^((r - q) × T)
F = 100 × e^((0.05 - 0.02) × 1)
F = 100 × e^(0.0300)
F = 100 × 1.030455

Calculated Futures Price (F): $103.05
```

Step 3: Display Futures Price and Analyze Relationship

```
print("\n" + "="*60)
print("STEP 3 & 4: Analyze Relationship Between Spot and Futures
Price")
print("="*60)

# Compare Spot Price and Futures Price
print(f"\nSpot Price (S): ${S:.2f}")
print(f"Futures Price (F): ${F:.2f}")
print(f"Difference (F - S): ${F - S:.2f}")
print(f"Percentage Change: {((F - S) / S * 100):.2f}%")

# Determine Market Condition
print("\n" + "-"*60)
if F > S:
    condition = "CONTANGO"
    explanation = "The market is in CONTANGO (positive cost of
carry)."
    details = "This means the futures price is higher than the spot
price,"
    details2 = "indicating that the cost of holding the asset
(interest rate)"
    details3 = "exceeds the income yield (dividends)."
elif F < S:
    condition = "BACKWARDATION"
    explanation = "The market is in BACKWARDATION (negative cost of
carry)."
    details = "This means the futures price is lower than the spot
price,"
    details2 = "indicating that the income yield exceeds the cost of
holding"
    details3 = "the asset."
else:
```

```
    condition = "FAIR VALUE"
    explanation = "The market is at FAIR VALUE."
    details = "The futures price equals the spot price,"
    details2 = "indicating the cost of carry equals the income yield."
    details3 = ""

print(f"Market Condition: {condition}")
print(explanation)
print(details)
if details2:
    print(details2)
if details3:
    print(details3)
```

```
============================================================
STEP 3 & 4: Analyze Relationship Between Spot and Futures Price
============================================================

Spot Price (S): $100.00
Futures Price (F): $103.05
Difference (F - S): $3.05
Percentage Change: 3.05%

------------------------------------------------------------
Market Condition: CONTANGO
The market is in CONTANGO (positive cost of carry).
This means the futures price is higher than the spot price,
indicating that the cost of holding the asset (interest rate)
exceeds the income yield (dividends).
```

Step 5: Sensitivity Analysis

```
print("\n" + "="*60)
print("STEP 5: Sensitivity Analysis")
print("="*60)

# 5.1: Vary Risk-Free Rate and observe changes in Futures Price
print("\n5.1: Effect of Risk-Free Rate on Futures Price")
print("-"*60)

interest_rates = np.arange(0.01, 0.11, 0.01)  # 1% to 10% in 1%
increments
futures_prices_by_rate = []

print(f"\n{'Interest Rate':<15} {'Futures Price':<15}
{'Difference':<15} {'% Change':<15}")
print("-"*60)

for rate in interest_rates:
```

```
    F_rate = S * np.exp((rate - q) * T)
    futures_prices_by_rate.append(F_rate)
    diff = F_rate - S
    pct_change = (diff / S) * 100
    print(f"{rate*100:>6.1f}%{'':<7} ${F_rate:>10.2f}{'':<3} $
{diff:>8.2f}{'':<5} {pct_change:>6.2f}%")

# 5.2: Vary Time to Maturity and observe changes in Futures Price
print("\n\n5.2: Effect of Time to Maturity on Futures Price")
print("-"*60)

time_periods = np.arange(0.1, 3.1, 0.2)  # 0.1 to 3 years in 0.2 year
increments
futures_prices_by_time = []

print(f"\n{'Time to Maturity':<20} {'Futures Price':<15}
{'Difference':<15} {'% Change':<15}")
print("-"*60)

for time in time_periods:
    F_time = S * np.exp((r - q) * time)
    futures_prices_by_time.append(F_time)
    diff = F_time - S
    pct_change = (diff / S) * 100
    print(f"{time:>6.1f} years{'':<8} ${F_time:>10.2f}{'':<3} $
{diff:>8.2f}{'':<5} {pct_change:>6.2f}%")
```

```
============================================================
STEP 5: Sensitivity Analysis
============================================================

5.1: Effect of Risk-Free Rate on Futures Price
------------------------------------------------------------

Interest Rate   Futures Price   Difference      % Change
------------------------------------------------------------
   1.0%        $     99.00    $   -1.00       -1.00%
   2.0%        $    100.00    $    0.00        0.00%
   3.0%        $    101.01    $    1.01        1.01%
   4.0%        $    102.02    $    2.02        2.02%
   5.0%        $    103.05    $    3.05        3.05%
   6.0%        $    104.08    $    4.08        4.08%
   7.0%        $    105.13    $    5.13        5.13%
   8.0%        $    106.18    $    6.18        6.18%
   9.0%        $    107.25    $    7.25        7.25%
  10.0%        $    108.33    $    8.33        8.33%


5.2: Effect of Time to Maturity on Futures Price
```

```
------------------------------------------------------------

Time to Maturity     Futures Price   Difference      % Change
------------------------------------------------------------
   0.1 years         $    100.30    $    0.30        0.30%
   0.3 years         $    100.90    $    0.90        0.90%
   0.5 years         $    101.51    $    1.51        1.51%
   0.7 years         $    102.12    $    2.12        2.12%
   0.9 years         $    102.74    $    2.74        2.74%
   1.1 years         $    103.36    $    3.36        3.36%
   1.3 years         $    103.98    $    3.98        3.98%
   1.5 years         $    104.60    $    4.60        4.60%
   1.7 years         $    105.23    $    5.23        5.23%
   1.9 years         $    105.87    $    5.87        5.87%
   2.1 years         $    106.50    $    6.50        6.50%
   2.3 years         $    107.14    $    7.14        7.14%
   2.5 years         $    107.79    $    7.79        7.79%
   2.7 years         $    108.44    $    8.44        8.44%
   2.9 years         $    109.09    $    9.09        9.09%
```

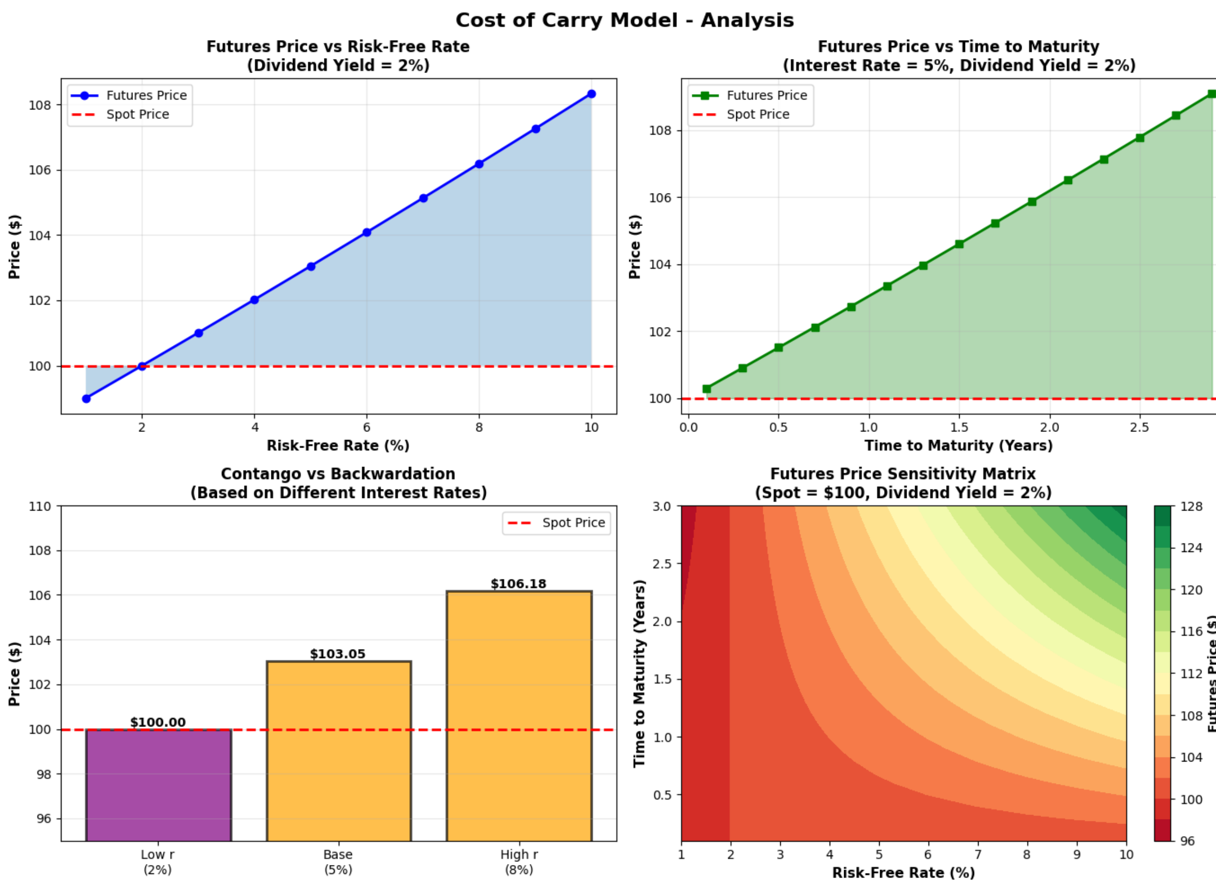
Step 6: Plot Graphs

```
print("\n" + "="*60)
print("STEP 6: Plot Graphs")
print("="*60)

fig, axes = plt.subplots(2, 2, figsize=(14, 10))
fig.suptitle('Cost of Carry Model - Analysis', fontsize=16,
fontweight='bold')

# Graph 1: Futures Price vs Risk-Free Rate
ax1 = axes[0, 0]
ax1.plot(interest_rates * 100, futures_prices_by_rate, 'b-o',
linewidth=2, markersize=6, label='Futures Price')
ax1.axhline(y=S, color='r', linestyle='--', linewidth=2, label='Spot
Price')
ax1.fill_between(interest_rates * 100, S, futures_prices_by_rate,
alpha=0.3)
ax1.set_xlabel('Risk-Free Rate (%)', fontsize=11, fontweight='bold')
ax1.set_ylabel('Price ($)', fontsize=11, fontweight='bold')
ax1.set_title('Futures Price vs Risk-Free Rate\n(Dividend Yield =
2%)', fontsize=12, fontweight='bold')
ax1.grid(True, alpha=0.3)
ax1.legend(fontsize=10)

# Graph 2: Futures Price vs Time to Maturity
ax2 = axes[0, 1]
ax2.plot(time_periods, futures_prices_by_time, 'g-s', linewidth=2,
markersize=6, label='Futures Price')
ax2.axhline(y=S, color='r', linestyle='--', linewidth=2, label='Spot
```

```
Price')
ax2.fill_between(time_periods, S, futures_prices_by_time, alpha=0.3,
color='green')
ax2.set_xlabel('Time to Maturity (Years)', fontsize=11,
fontweight='bold')
ax2.set_ylabel('Price ($)', fontsize=11, fontweight='bold')
ax2.set_title('Futures Price vs Time to Maturity\n(Interest Rate = 5%,
Dividend Yield = 2%)', fontsize=12, fontweight='bold')
ax2.grid(True, alpha=0.3)
ax2.legend(fontsize=10)

# Graph 3: Contango/Backwardation Comparison
ax3 = axes[1, 0]
scenarios = ['Low r\n(2%)', 'Base\n(5%)', 'High r\n(8%)']
scenario_rates = [0.02, 0.05, 0.08]
scenario_futures = [S * np.exp((rate - q) * T) for rate in
scenario_rates]
colors = ['orange' if f > S else 'purple' for f in scenario_futures]
bars = ax3.bar(scenarios, scenario_futures, color=colors, alpha=0.7,
edgecolor='black', linewidth=2)
ax3.axhline(y=S, color='r', linestyle='--', linewidth=2, label='Spot
Price')
ax3.set_ylabel('Price ($)', fontsize=11, fontweight='bold')
ax3.set_title('Contango vs Backwardation\n(Based on Different Interest
Rates)', fontsize=12, fontweight='bold')
ax3.set_ylim([95, 110])
ax3.legend(fontsize=10)
ax3.grid(True, alpha=0.3, axis='y')

# Add value labels on bars
for bar, price in zip(bars, scenario_futures):
    height = bar.get_height()
    ax3.text(bar.get_x() + bar.get_width()/2., height,
             f'${price:.2f}',
             ha='center', va='bottom', fontweight='bold', fontsize=10)

# Graph 4: Sensitivity Matrix (Heatmap-style)
ax4 = axes[1, 1]
rates_range = np.arange(0.01, 0.11, 0.01)
time_range = np.arange(0.1, 3.1, 0.1)
sensitivity_matrix = np.zeros((len(time_range), len(rates_range)))

for i, t in enumerate(time_range):
    for j, rate in enumerate(rates_range):
        sensitivity_matrix[i, j] = S * np.exp((rate - q) * t)

im = ax4.contourf(rates_range * 100, time_range, sensitivity_matrix,
levels=15, cmap='RdYlGn')
cbar = plt.colorbar(im, ax=ax4)
cbar.set_label('Futures Price ($)', fontweight='bold')
```

```
ax4.set_xlabel('Risk-Free Rate (%)', fontsize=11, fontweight='bold')
ax4.set_ylabel('Time to Maturity (Years)', fontsize=11,
fontweight='bold')
ax4.set_title('Futures Price Sensitivity Matrix\n(Spot = $100,
Dividend Yield = 2%)', fontsize=12, fontweight='bold')

plt.tight_layout()
plt.show()

print("\nGraphs plotted successfully!")
```

```
============================================================
STEP 6: Plot Graphs
============================================================
```




```
Graphs plotted successfully!
```

Conclusions and Key Findings

```
print("\n" + "="*60)
print("CONCLUSIONS AND KEY FINDINGS")
print("="*60)

conclusions = """
1. COST OF CARRY MODEL VALIDITY
   • The Cost of Carry Model successfully determines the theoretical
futures price
     based on spot price, interest rate, dividend yield, and time to
maturity.
   • Formula F = S × e^((r - q) × T) accurately reflects the pricing
relationship.

2. MARKET CONDITIONS (CONTANGO vs BACKWARDATION)
   • When the interest rate (r) exceeds dividend yield (q), F > S
(CONTANGO)
     - The market reflects the cost of carrying the asset until
maturity
     - Investors expect to pay more for future delivery due to
financing costs

   • When dividend yield (q) exceeds interest rate (r), F < S
(BACKWARDATION)
     - The income generated from the asset offsets carrying costs
     - Future prices become cheaper than spot prices

3. SENSITIVITY TO INTEREST RATES
   • Direct Relationship: As interest rates increase, futures prices
increase
   • 1% increase in interest rate results in a proportional increase
in futures price
   • This is critical for hedging decisions in fixed-income and
commodity markets

4. SENSITIVITY TO TIME TO MATURITY
   • Non-linear Relationship: Longer maturity periods amplify the
effect of cost of carry
   • Greater time = Greater accumulated carrying costs
   • Time value of money significantly impacts pricing in long-dated
contracts

5. PRACTICAL IMPLICATIONS
   • Arbitrage Opportunities: If F ≠ S × e^((r - q) × T), profit
opportunities exist
   • Portfolio Hedging: Investors can use this model to optimize
hedging strategies
   • Market Efficiency: Deviations from the model indicate market
inefficiencies
```

```
   • Risk Management: Understanding cost of carry helps in managing
commodity risk

6. ASSUMPTIONS & LIMITATIONS
   • Assumes continuous compounding and no transaction costs
   • Assumes constant interest rates and dividend yields
   • Real markets may have storage costs, convenience yields, and
transaction frictions
   • Model works best for commodities with predictable storage and
yield characteristics

7. KEY TAKEAWAYS
   ✓ The Cost of Carry Model is fundamental to modern futures pricing
theory
   ✓ Understanding F = S × e^((r - q) × T) is essential for commodity
traders
   ✓ Deviations from theoretical prices signal market opportunities
   ✓ Sensitivity analysis reveals how external factors affect futures
contracts
   ✓ Model forms the basis for advanced derivative pricing models
"""

print(conclusions)

# Summary Statistics
print("\n" + "="*60)
print("SUMMARY STATISTICS FROM ANALYSIS")
print("="*60)

summary_df = pd.DataFrame({
    'Metric': [
        'Base Spot Price',
        'Base Futures Price',
        'Base Cost of Carry',
        'Min Interest Rate',
        'Max Interest Rate',
        'Min Futures Price (at 1%)',
        'Max Futures Price (at 10%)',
        'Min Time Period',
        'Max Time Period',
        'Min Futures (0.1 years)',
        'Max Futures (3.0 years)'
    ],
    'Value': [
        f'${S:.2f}',
        f'${F:.2f}',
        f'{(r-q)*100:.2f}%',
        '1%',
        '10%',
        f'${S * np.exp((0.01 - q) * T):.2f}',
```

```
        f'${S * np.exp((0.10 - q) * T):.2f}',
        '0.1 years',
        '3.0 years',
        f'${S * np.exp((r - q) * 0.1):.2f}',
        f'${S * np.exp((r - q) * 3.0):.2f}'
    ]
})

print(summary_df.to_string(index=False))
print("\n" + "="*60)
print("Lab 10 - Cost of Carry Model Implementation Complete!")
print("="*60)
```

```
============================================================
CONCLUSIONS AND KEY FINDINGS
============================================================

1. COST OF CARRY MODEL VALIDITY
   • The Cost of Carry Model successfully determines the theoretical
futures price
     based on spot price, interest rate, dividend yield, and time to
maturity.
   • Formula F = S × e^((r - q) × T) accurately reflects the pricing
relationship.

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maturity
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```

```
   • Greater time = Greater accumulated carrying costs
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   • Arbitrage Opportunities: If F ≠ S × e^((r - q) × T), profit
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traders
   ✓ Deviations from theoretical prices signal market opportunities
   ✓ Sensitivity analysis reveals how external factors affect futures
contracts
   ✓ Model forms the basis for advanced derivative pricing models


============================================================
SUMMARY STATISTICS FROM ANALYSIS
============================================================
                    Metric     Value
           Base Spot Price   $100.00
        Base Futures Price   $103.05
        Base Cost of Carry     3.00%
         Min Interest Rate        1%
         Max Interest Rate       10%
 Min Futures Price (at 1%)    $99.00
Max Futures Price (at 10%)   $108.33
           Min Time Period 0.1 years
           Max Time Period 3.0 years
   Min Futures (0.1 years)   $100.30
   Max Futures (3.0 years)   $109.42

============================================================
```

```
Lab 10 - Cost of Carry Model Implementation Complete!
============================================================
```